Precedent

Every incident resolved becomes precedent.

Team — UK AI Agent Hackathon EP5 · Conduct “Make Legacy Move” · Fetch.ai · BasedAI

Something breaks → find the manual → click through an admin console.

8.85 hours

average time to resolve an incident — for a fix that is often minutes of keystrokes once it's known

When Sky News went off air in the CrowdStrike outage, the fix was a documented admin procedure executed thousands of times by hand — 8.85 business hours of queueing and lookup, not fixing (MetricNet/HDI).

$600B / year

downtime cost across the Global 2000 — up 50% in two years (Splunk, 2026)

≈ $200M per company (Oxford Economics/Splunk, 2024)

>60% of incidents are repeats — the fix already exists, nobody can find it (ServiceNow’s own support org)

Downtime costs the Global 2000 ~$600B/yr — about $200M per company — and at ServiceNow’s own support desk, >60% of incidents were repeats whose fix already existed but couldn’t be found.

AI SREs fix broken code. In real enterprises, the fix is almost never code — it's a documented admin change.

Precedent remembers every fix your organisation has ever applied — and applies it again.

risk-classified · approval-gated · audited · reversible

detect → find the documented fix → check risk → get approval → execute → verify → remember

The memory unit is not a document — it's an executed fix with provenance (symptom, verified fix, approver, risk class, rollback, outcome). A human approval gate, a full audit trail, and a rollback prepared before anything runs.

Same incident. Same fix.

The second time is free.

Three bars to scale: the manual loop is 8.85 hours of queueing and searching; Precedent’s first resolution ~60 s (a human still clicks approve); every repeat after ~15 s. The 15 s fast-path skips LLM triage for a known class — engineered, not hoped-for.

MediaCo — a simulated broadcaster · real data · live Jira Service Management · unscripted inputs

1. Broken EPG publish → plan + rollback → human approves → fixed in ~60 s, Jira ticket closes itself
2. **Same class recurs → pre-approved standard change → fixed in ~15 s**
3. Rights conflict, restricted runbook → Precedent refuses and routes to the right team

Seed data: real public runbooks (GitLab, Kubernetes SIG, the published CrowdStrike remediation), real programme metadata, and a 141k-event public IT-incident log ingested as fix history. Incident text is mutated at generation time — typos, vague symptoms, missing codes.

The incident text is deliberately mangled — typos, wrong terminology, missing codes — and any judge can file a ticket live. Incident 3: it isn't allowed to read that runbook, so it refused and routed to the team that is. It knows what it's not allowed to touch.

Autonomy is earned per incident-class — and a human grants it.

L0 Observe → L1 Recommend → L2 Execute after approval → **L3 Standing Approval** (pre-approved standard change)

Promote to standing approval

Revoke

Approval moves earlier in time. It never leaves the loop.

No one approved that second ticket — the operations lead pre-approved the fix class after watching it succeed. Standing approval is an ITIL standard change: earned through verified history, revocable with one click, demoted automatically on any failure.

Five specialised agents · deterministic policy engine · permission-aware memory

Live Jira Service Management · open-weight models end-to-end · agents live on Fetch.ai Agentverse / ASI:One

Five specialised agents run the loop; a deterministic policy engine — not the model — decides what may execute; and the memory inherits the permissions of the documents and tickets it learned from, live-synced from Jira. (Q&A-only on stage.)

AI SREs stop at diagnosis. RPA is incident-blind. Runbook automation only runs pre-written scripts. ServiceNow deflects tickets inside its own walls.

The closed loop — retrieve your documented fix → approval-gated execution in third-party systems → verified, audited, remembered — is unclaimed.

ServiceNow paid $2.85B for Moveworks — resolution memory is worth buying.

$340M+ of AI-SRE VC since 2024 points at code/infra diagnosis and stops at 'here's the likely cause.' Komodor/Ignio execute in their own silos — the whitespace is the combination: your documented fix + business-app execution + graduated approval + compounding memory. (Q&A-only on stage.)

Every resolution writes a record: symptom → verified fix → approver → risk → rollback → outcome.

incident → fix → precedent → faster next time → more incidents trusted to it → ↻

141,000 real incident events ingested · a 25k-record incident store

94% arrived with their fix already precedented — and still took a median 18 hrs (calendar) to resolve by hand

Permission-checked retrieval: P99 0.445 μs · FNR 0 leaks / 5,219 deny-expected · 6/6 adversarial attacks

Triage on mangled tickets (100-mutation corpus): 0 false-fast-paths / 100 · 25/25 red-herring decoys resisted (8% correct-match, 50% safe-degrade, 0% false fast-path)

Documents go stale; executed fixes with provenance don't. Every incident Precedent resolves makes the next one faster — accuracy becomes a function of tenure in the account. 94% precedent existence is measured on 24,918 real UCI incidents; 18 hrs is calendar, never blended with the 8.85 business-hour baseline. (Q&A-only on stage.)

Media ops first — regulated failure (Ofcom), 24/7 deadlines, legacy consoles

One MSP deal = hundreds of channels — Encompass runs 1,200+ channels/day; Amagi 5,000+ deliveries (vendor-claimed)

Then: every ops team drowning in runbooks — every ticket deflected saves $22 → $69 → $104 per escalation tier

Agents made retrieve-and-safely-execute possible this year. Incumbents are priced to meter the workflow, not delete it.

Media first: first-hand pain, regulator-documented failures, Netflix proved the pattern in-house. The MSP channel is the wedge-scaler — fixed-price SLA ops means every auto-remediated incident is pure margin. $22/$69/$104 = MetricNet cost-per-ticket (vintage ~2019–20).

Founding team — UK AI Agent Hackathon EP5

(one of us watched the incident loop first-hand inside broadcast-streaming operations)

The ask: two intros to broadcast-ops or MSP design partners — and we’re applying to Antler & EWOR with this.

ServiceNow paid $2.85B for resolution memory. We’re building the version that executes.

The ask, as a sentence: two introductions to anyone running a 24/7 ops or NOC team — media is where we start, not where we stop — and we’re applying to Antler and EWOR. (Team names/photos are added by the founders before the live pitch.)

A1 — Every number, sourced

Global 2000 downtime bill **$600B/yr, +50% in 2 yrs** Splunk 2026 (exact framing)
Per-company downtime cost **$200M/yr** Oxford Economics/Splunk 2024 — verified 3–0
Repeat incidents with existing fixes **>60%** ServiceNow KCS case study — verified 3–0
Cost per ticket L1 → desktop → L3 **$22 → $69 → $104** MetricNet whitepaper (~2019–20)
Average incident MTTR **8.85 business hrs** MetricNet/HDI
KB-attached / KCS relief **66% faster / 52% faster** ServiceNow — verified
Moveworks acquisition **$2.85B** ServiceNow newsroom, Mar 2025
Deflection precedent **50–88%** Moveworks — vendor-claimed
Fix-class match rate (24,918 real incidents) **94.4% (symptom 98.6%)** our measurement, UCI CC BY 4.0
Median resolution, precedented repeats **18.2 calendar hrs (p75 136.6)** ours — never blend w/ 8.85 business
Recurring classes ≥4 occurrences **558 classes = 94.8% of volume** ours — ladder-bootstrap evidence
Extractor robustness (100-mutation corpus) **0 false-fast-paths / 100; 25/25 decoys resisted** ours — seed 4207
Conformance FNR / attacks **0 leaks / 5,219; 6/6** ours — independent-oracle bench

A2 — Data provenance (the “not made up” slide)

KB seeded with real published runbooks (GitLab, Kubernetes SIG, the CrowdStrike channel-file remediation) — every article carries its adapted_from URL.

Fix history from the UCI ServiceNow incident event log (141,712 events / 24,918 incidents, CC BY 4.0) — measured 94% fix-class match, 18.2h calendar median for precedented repeats.

Scheduler/rights/EPG seeded with real UK programme metadata (TVmaze CC BY-SA / Freeview XMLTV) and real streaming-catalog data (CC0 Kaggle) — only licence-window terms are synthesised, by a stated rule.

We checked licences and REJECTED TMDB (API terms prohibit AI/ML use) and IMDb — the diligence is itself a credibility beat.

Incident text mutated at generation time (typos, colloquial symptoms, missing codes, red herrings). The systems are simulated; the content is real.

A3 — Standing approval: exact semantics

Incident-class key = deterministic fingerprint sha256(service | error_code | target_object_type) from structured fields — the LLM proposes candidates, but a match only counts on fingerprint equality. The model never authorises itself.

Graduation, printed in the audit log: 3 consecutive verified L2 successes, zero rollbacks → eligible; a human clicks Promote to Standing Approval. Any verification failure or rollback auto-demotes the class to L1 and logs a demotion event.

The rollback plan is generated BEFORE execution and is a precondition of the gate; verification failure fires it automatically.

Approver identity is recorded per action (chat: the Chat Protocol sender address; production: SSO identity — the control structure is identical).

A4 — Permission-aware memory (BasedAI depth)

Every record stores the FULL set of source permission constraints; retrieval must satisfy ALL of them — conjunction, not one strictest label.

A precompiled effective-policy cache makes the check one indexed lookup: P99 0.445 µs (end-to-end overhead P99 0.0130 ms) over the 25k-record store, concurrent queries.

Benchmarked in the track's own vocabulary: FNR 0 / 5,219 and FPR 0 / 4,781 over 10,000 ground-truth queries graded by an INDEPENDENT oracle, P50/P99, overhead, ACL drift, time-to-consistency — each vs threshold in bench/RESULTS.md; six named attacks pass.

Revocation cascades — dual enforcement: flip the issue security level in Jira → Jira hides the runbook AND every derived fix/summary/embedding becomes unretrievable within one poll tick, with denial audit events in BOTH logs.

Fallback mode fails closed: uncertain ACL freshness → not served. Ships as a standalone library; 100% open-weight models, named in the README.

A5 — Why won’t ServiceNow / Conduct build this?

ServiceNow monetises seats + tickets; auto-resolution cannibalises both. They ACQUIRE this layer: Jeli $29.7M, Moveworks $2.85B. We’re building what they buy.

Now Assist acts inside ServiceNow workflows; we execute inside third-party business-app admin surfaces — their walls are our territory.

Conduct makes legacy legible; we make it operable — a complement, and the brief told us to build exactly this (“start with the process”).

Rehearsed: “They meter the workflow; we delete it. Incumbents don’t ship products that shrink their own ticket count — they buy them. That’s the Moveworks story.”

A6 — Integration reality (the “you built the simulator” defence)

Four-tier surface strategy: REST APIs (WhatsOn-class schedulers) → BXF/file exchange → watched-folder/FTP drops → RPA-style console driving (last resort, same approval gates).

The Operator executes only TYPED tool calls per surface — never free-form shell; tier 3/4 fixes start at L0/L1 and stay approval-gated.

MediaCo’s endpoints deliberately mirror the surfaces the real vendors document (Grass Valley STRATUS: “traffic integration via BXF files”).

A7 — Fetch.ai deliverables

Three agents registered on Agentverse as mailbox agents, collaborating over the uAgents Agent Chat Protocol; both Innovation Lab + hackathon badges on each profile; the hosted degraded-L0 Watcher stays live post-hackathon.

The full loop runs inside an ASI:One conversation: report an incident in chat → plan + rollback → “approve” → the ticket transitions and closes → audit hash returned.

Approval principal = the Chat Protocol sender address, logged as the authorising identity, with a 10-minute gate TTL (a dropped session never leaks an execution).

Agentverse profile URLs + the ASI:One shared-chat URL are captured at live registration.

A9 — Bottoms-up ACV

One arithmetic slide, every input sourced:

1 MSP NOC (Encompass-class, 1,200+ channels) × tens of thousands of tickets/yr × 60%+ repeat share (ServiceNow, verified 3–0; our corpus says 94% precedented) × $50 blended deflection value (inside the $22–$104 MetricNet ladder)

⇒ seven-figure ACV justification per site — before counting downtime avoidance.

This is the bottoms-up serviceable-market answer; it does not replace primary customer evidence and never pretends to.

A8 — Liability & regulated ops

Precedent executes only documented, previously-verified fixes at the customer’s chosen autonomy level — the customer’s own change-management policy, encoded and enforced. Shadow-mode-first at MSPs.

Headline metric we quote: per-class execution success + rollback rate from real runs — never model benchmarks (the incident.io-vs-Rootly accuracy-theatre war is the cautionary tale).

Every action carries: input, reasoning, confidence, action, authorising rule + approver — the audit schema regulators and SOC 2 auditors already recognise.

What exists Monday morning

Durable artifacts — not slideware:

- A hosted degraded-L0 Watcher, live on Fetch.ai Agentverse (answers a described incident at L0, never executes).
- precedent_memory — the permission-aware memory as a standalone, importable Python library (fail-closed retrieval, hash-chained audit, conformance bench).
- The ground-truth conformance bench, graded by an independent oracle (FNR 0 / 5,219, 6/6 attacks, seed 4207 — byte-identically reproducible).
- The public evidence index (docs/evidence, docs/compliance) — every claim traceable to a committed artifact.